

Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov/ Dec -2024

SOFT COMPUTING TECHNIQUES [ELE-4062]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Write the definition of the following activation function along with their graphical representation.
- A) a. Binary Step
b. ReLU
c. Leaky ReLU
d. Sigmoid
e. Bipolar Sigmoid (5)
- B) Two fuzzy sets $\tilde{A}$ and $\tilde{B}$ is defined as $\tilde{A} = \left\{ \frac{0.2}{10} + \frac{1}{12} + \frac{0.3}{14} \right\}$ and $\tilde{B} = \left\{ \frac{0.2}{3} + \frac{1}{6} + \frac{0.4}{9} \right\}$. Write the extension principle for binary operation between the fuzzy sets $\tilde{A}$ and $\tilde{B}$. Also, carry out the following arithmetic operations between $\tilde{A}$ and $\tilde{B}$. (5)
- a. $\tilde{A} + \tilde{B}$
b. $\tilde{A} - \tilde{B}$
- 2) (5)
- A)

Consider a fuzzy logic controller is used to control the speed of a motor by changing its input voltage (V) according to two input variables; speed (SP), and speed change rate SC. Let the fuzzy set of SP be {Slow (S), Normal (N), Fast (F)}, and the fuzzy set for SC be {Low (L), Medium (M), High (H)}, and for the control action be {Slow Down (DN), No Change (NC), Speed Up (Up)}, where, $(SP \in [500, 1000])$, $(SC \in [0, 10])$, and $(V \in [2, 3])$ with step = 0.1. The membership functions for the input/output variables are described below.

SP		SC		V	
Fuzz y term	Membership function	Fuzz y term	Membershi p function	Fuzzy term s	Membership function
S	$L(x; 600, 750)$	L	$L(y; 2, 4)$	DN	$L(u; 2.2, 2.5)$
N	$\Lambda(x; 600, 750, 900)$	M	$\pi(y; 2, 4, 6, 8)$	NC	$\Lambda(u; 2.2, 2.5, 2.6)$
F	$\Gamma(x; 750, 900)$	H	$\Gamma(y; 6, 8)$	Up	$\Gamma(u; 2.5, 2.8)$

The fuzzy rules are shown below.

Λ	S	N	F
L	Up	NC	NC
M	Up	NC	NC
H	NC	DN	DN

Find the rules with non-zero degree of firing and their degree of firing corresponding to the control action SP=910 And SC= 6.5. Also, find the control action if SP = 910 And SC = 6.5. Use the mean of the maximum technique for defuzzification.

- B) The control action of a physical system is characterized by the following four fuzzy rules.

R_1 : If x is small and y is small then $z = x - y + 1$

R_2 : If x is small and y is large then $z = y + 3$

R_3 : If x is large and y is small then $z = x - 3$

R_4 : If x is large and y is large then $z = 2x + y + 2$

(5)

Find the rules with non-zero degree of firing and their degree of firing corresponding to the control action if $x = 6$ and $y = 5$. Also, Find the control action if $x = 6$ and $y = 5$ using Takagi-Sugano model. *Small* and *large* are two fuzzy numbers given by $\Lambda(x; 2, 5, 8)$ and $\Lambda(y; 5, 7, 8)$.

- 3) What does gene and allele mean in genetic algorithm. (2)

- A)
- B) Discuss the one point and multi point crossover process of genetic algorithms with suitable example. (4)
- C) Discuss bit flip and swap mutation process of genetic algorithm with suitable example. (4)
- 4) What are the terms genotype and phenotype mean in genetic algorithm. (2)
- A)
- B) Describe the steps involved in the genetic algorithm with a flow chart. (3)
- C) Discuss the Roulette wheel selection process with example of five chromosomes A, B, C, D, and E with fitness values 8, 3, 1, 2, and 6 respectively. Also find the assigned angles with each chromosomes A, B, C, D, and E. (5)
- 5) Define a fuzzy set. (2)
- A)
- B) Discuss the difference between crisp/classical set and fuzzy set with the help of membership function of the crisp and fuzzy set. (3)
- C) Consider two fuzzy relations $\tilde{R}$ and $\tilde{S}$ given as follows:

$$\tilde{R} = \begin{matrix} & -100 & -50 & 0 & 50 & 100 \\ \begin{matrix} 9 \\ 18 \\ 27 \\ 36 \end{matrix} & \begin{bmatrix} 0.2 & 0.5 & 0.7 & 1 & 0.9 \\ 0.3 & 0.5 & 0.7 & 1 & 0.8 \\ 0.4 & 0.6 & 0.8 & 0.9 & 0.4 \\ 0.9 & 1 & 0.8 & 0.6 & 0.4 \end{bmatrix} \end{matrix} \quad \tilde{S} = \begin{matrix} & 2 & 4 & 8 & 16 & 20 \\ \begin{matrix} -100 \\ -50 \\ 0 \\ 50 \\ 100 \end{matrix} & \begin{bmatrix} 1 & 0.8 & 0.6 & 0.3 & 0.1 \\ 0.7 & 1 & 0.7 & 0.5 & 0.4 \\ 0.5 & 0.6 & 1 & 0.8 & 0.8 \\ 0.3 & 0.4 & 0.6 & 1 & 0.9 \\ 0.9 & 0.3 & 0.5 & 0.7 & 1 \end{bmatrix} \end{matrix} \quad (5)$$

If $\tilde{R}$ is relationship between frequency and temperature, and $\tilde{S}$ represents a relation between temperature and reliability index of circuit. Obtain the relationship between frequency and reliability using max-product composition.

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